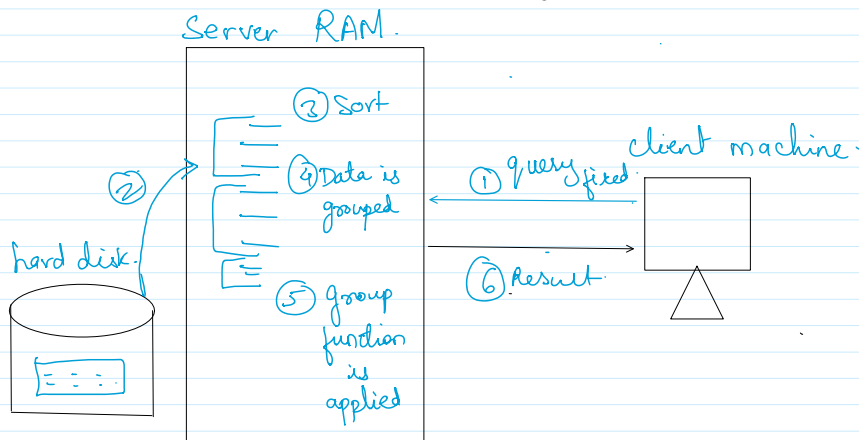
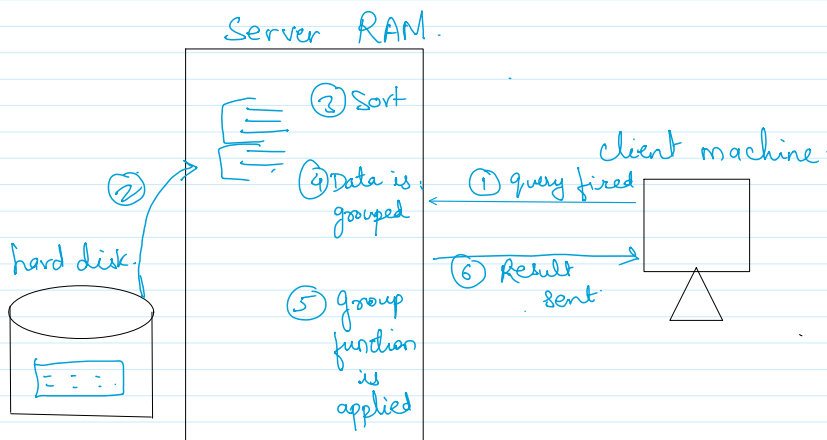


```
SELECT job, COUNT(empno)
FROM emp
GROUP BY job;
```



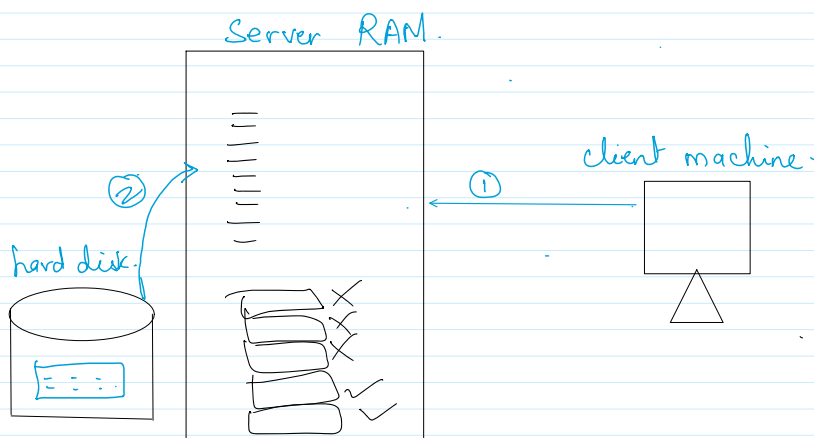
- 1) Query is Fired by the client to the server.
- 2) Server Fetches all the rows from the hard disk into the RAM as there is no WHERE clause.
- 3) The data is sorted in the RAM on the basis of the grouped column (This helps to group similar rows together).
- 4) Data is grouped on the basis of job.
- 5) Group function is applied on each group.
- 6) Result is sent back to the client.

```
SELECT job, COUNT(empno)
FROM emp
WHERE job IN('manager', 'analyst')
GROUP BY job;
```



- 1) Query is Fired by the client to the server.
- 2) Server Fetches only the rows which satisfy the where clause from the hard disk into the RAM.
- 3) The data is sorted in the RAM on the basis of the grouped column (This helps to group similar rows together).
- 4) Data is grouped.
- 5) Group function is applied on each group (In this there are only 2 groups, manager and analyst).
- 6) Result is sent back to the client.

```
SELECT job, COUNT(empno)
FROM emp
GROUP BY job
HAVING job IN('manager', 'analyst');
```



- 1) Query is Fired by the client to the server.
- 2) Server Fetches all the rows from the hard disk into the RAM as there is no WHERE clause.
- 3) The data is sorted in the RAM on the basis of the grouped column (This helps to group similar rows together).
- 4) Data is grouped.
- 5) Group function is applied on each group. There will be 5 groups. Having clause will send only 2 groups to the client as output. But here there was unnecessary working on all the groups.

